B5 General Relativity — Model Solutions, Trinity 2021

Question 1 — The falling apple in General Relativity

Problem. An apple of unit mass falls a height h from a tree to the ground in vacuum. Treat the problem fully in GR: derive the geodesic equation, choose the metric, evaluate the connection coefficients, show the motion is purely radial, find $r(t)$ as seen by a nearby observer, and use a conserved energy to obtain $\frac{1}{2}(\mathrm{d}r/\mathrm{d}\tau)^2 = GM[1/R_E - 1/(R_E + h)]$.

One-sentence statement. Gravity is the curvature of spacetime sourced by stress–energy; free particles follow timelike geodesics of the metric.

(a) Geodesic equation from local inertial coordinates

In the apple's local inertial frame (LIF), ξ^α are Riemann normal coordinates and τ is the proper time read off the apple's worldline ($c^2\,\mathrm{d}\tau^2 = -\eta_{\alpha\beta}\,\mathrm{d}\xi^\alpha\,\mathrm{d}\xi^\beta$). Free fall:

$$\frac{\mathrm{d}^2\xi^\alpha}{\mathrm{d}\tau^2} = 0. \tag{1}$$

Apply the chain rule to $x^\mu(\xi)$:

$$\frac{\mathrm{d}\xi^\alpha}{\mathrm{d}\tau} = \frac{\partial\xi^\alpha}{\partial x^\mu}\,\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}.$$

Differentiate once more in τ:

$$\frac{\mathrm{d}^2\xi^\alpha}{\mathrm{d}\tau^2} = \frac{\partial\xi^\alpha}{\partial x^\mu}\,\frac{\mathrm{d}^2x^\mu}{\mathrm{d}\tau^2} + \frac{\partial^2\xi^\alpha}{\partial x^\mu\partial x^\nu}\,\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\,\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau}.$$

Set the LHS to zero by (**??**):

$$\frac{\partial\xi^\alpha}{\partial x^\mu}\,\frac{\mathrm{d}^2x^\mu}{\mathrm{d}\tau^2} + \frac{\partial^2\xi^\alpha}{\partial x^\mu\partial x^\nu}\,\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\,\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau} = 0.$$

Contract with $\partial x^\lambda/\partial\xi^\alpha$ and use $(\partial x^\lambda/\partial\xi^\alpha)(\partial\xi^\alpha/\partial x^\mu) = \delta^\lambda_\mu$:

$$\frac{\mathrm{d}^2x^\lambda}{\mathrm{d}\tau^2} + \Gamma^\lambda_{\mu\nu}\,\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\,\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau} = 0, \qquad \Gamma^\lambda_{\mu\nu} \equiv \frac{\partial x^\lambda}{\partial\xi^\alpha}\,\frac{\partial^2\xi^\alpha}{\partial x^\mu\partial x^\nu}.$$

Specialise to the instant the apple is at rest in x^i: only $\mathrm{d}x^0/\mathrm{d}\tau = c\,\mathrm{d}t/\mathrm{d}\tau$ survives the sum:

$$\boxed{\frac{\mathrm{d}^2x^\mu}{\mathrm{d}\tau^2} + c^2\Gamma^\mu_{00}\left(\frac{\mathrm{d}t}{\mathrm{d}\tau}\right)^2 = 0.} \tag{2}$$

Symbols: τ is proper time; x^μ are coordinates of the chosen chart; c converts coordinate time $t = x^0/c$; Γ^μ_{00} are Christoffel symbols of the metric in those coordinates.

(b) Choice of metric

The Earth is static, spherical, uncharged, vacuum exterior $\Rightarrow$ Schwarzschild solution (Birkhoff). Schwarzschild line element:

$$\mathrm{d}s^2 \;=\; -\Big(1-\frac{r_s}{r}\Big)\,c^2\mathrm{d}t^2 \;+\; \Big(1-\frac{r_s}{r}\Big)^{-1}\mathrm{d}r^2 \;+\; r^2(\mathrm{d}\theta^2+\sin^2\theta\,\mathrm{d}\phi^2), \tag{3}$$

with $r_s = 2GM/c^2$. Terms: t is the coordinate time of a distant inertial observer; r is the areal radius; (θ,ϕ) angular; M the Earth's mass.

Approximations for an apple near the Earth's surface:

- Weak field: $r_s/R_E \approx 1.4\times10^{-9} \ll 1$ at $r = R_E \approx 6.4\times10^6$ m.
- Slow motion: $|\mathrm{d}r/\mathrm{d}t|/c \ll 1$, so $\mathrm{d}\tau \approx \mathrm{d}t$ to lowest order in v/c and r_s/r.
- Static apple at $\tau = 0$: only Γ^{μ}_{00} matters in (**??**).
- Vacuum: $T_{\mu\nu} = 0$ outside the Earth, justifying Schwarzschild.

(c) Connection coefficients and radial motion

Schwarzschild metric is diagonal with $g_{00} = -c^2A(r)$, $g_{rr} = 1/A(r)$, $g_{\theta\theta} = r^2$, $g_{\phi\phi} = r^2\sin^2\theta$, where $A(r) = 1 - r_s/r$.

Use the static, diagonal-metric formula for Γ^{μ}_{00}:

$$\Gamma^{\mu}_{00} \;=\; -\tfrac{1}{2}g^{\mu\mu}\,\partial_\mu g_{00} \quad \text{(no sum)}.$$

Time component (metric static, $\partial_t g_{00} = 0$):

$$\Gamma^{0}_{00} \;=\; -\tfrac{1}{2}g^{00}\partial_0 g_{00} \;=\; 0.$$

Radial component, expand $g_{00} = -c^2A(r)$ and $g^{rr} = A(r)$:

$$\Gamma^{r}_{00} \;=\; -\tfrac{1}{2}A(r)\,\partial_r[-c^2A(r)].$$

Pull out the minus sign and the constant c^2:

$$\Gamma^{r}_{00} \;=\; \tfrac{c^2}{2}A(r)A'(r).$$

Differentiate $A(r) = 1 - r_s/r$ to give $A'(r) = r_s/r^2$:

$$\Gamma^{r}_{00} \;=\; \frac{c^2 r_s}{2r^2}\Big(1-\frac{r_s}{r}\Big).$$

Angular components (g_{00} independent of θ,ϕ):

$$\Gamma^{\theta}_{00} \;=\; 0, \qquad \Gamma^{\phi}_{00} \;=\; 0.$$

Insert into (**??**): only the $\mu = r$ component is non-zero:

$$\frac{\mathrm{d}^2r}{\mathrm{d}\tau^2} \;=\; -c^2\,\Gamma^{r}_{00}\Big(\frac{\mathrm{d}t}{\mathrm{d}\tau}\Big)^2 \;=\; -\frac{c^4 r_s}{2r^2}\Big(1-\frac{r_s}{r}\Big)\Big(\frac{\mathrm{d}t}{\mathrm{d}\tau}\Big)^2. \tag{4}$$

Hence the motion is purely radial. $\boxed{\mathrm{d}^2\theta/\mathrm{d}\tau^2 = \mathrm{d}^2\phi/\mathrm{d}\tau^2 = 0.}$

Choice of time. Nearby observer under the tree is essentially at rest at $r = R_E$ in the same Schwarzschild coordinates; their proper time is $\mathrm{d}\tau_{\text{obs}} = \sqrt{A(R_E)}\,\mathrm{d}t \approx \mathrm{d}t$ to one part in 10^9. Use coordinate time t as the observer's clock.

Weak-field, slow-motion limit of (**??**): drop r_s/r in the bracket and set $\mathrm{d}t/\mathrm{d}\tau \to 1$:

$$\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} \;\approx\; -\frac{c^2 r_s}{2r^2}.$$

Substitute $r_s = 2GM/c^2$:

$$\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} \;\approx\; -\frac{GM}{r^2}.$$

Newtonian recovery near the surface, set $r = R_E + z$ with $z \ll R_E$:

$$\frac{\mathrm{d}^2 z}{\mathrm{d}t^2} \;\approx\; -g, \qquad g \equiv GM/R_E^2.$$

Integrate once with $\dot{z}(0) = 0$:

$$\dot{z}(t) \;=\; -gt.$$

Integrate again with $z(0) = h$:

$$\boxed{r(t) \;=\; R_E + h - \tfrac{1}{2}gt^2.} \tag{5}$$

Approximations: weak field ($r_s/R_E \to 0$), slow motion ($v \ll c$), small fall height ($h \ll R_E$ so g constant), observer's proper time $\approx t$.

(d) Conserved energy and impact velocity

Schwarzschild has Killing vector ∂_t; for unit mass the conserved energy is

$$E \;=\; -g_{0\mu}\,\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau} \;=\; c^2 A(r)\,\frac{\mathrm{d}t}{\mathrm{d}\tau}. \tag{6}$$

Initial condition: stem snaps at $r = R_E + h$ with $\mathrm{d}r/\mathrm{d}\tau = 0$. Apply the 4-velocity normalisation $g_{\mu\nu}u^\mu u^\nu = -c^2$ at $\tau = 0$:

$$-c^2 A(R_E + h)\left(\frac{\mathrm{d}t}{\mathrm{d}\tau}\right)_0^2 \;=\; -c^2.$$

Solve for the initial $\mathrm{d}t/\mathrm{d}\tau$:

$$\left(\frac{\mathrm{d}t}{\mathrm{d}\tau}\right)_0 \;=\; A(R_E + h)^{-1/2}.$$

Substitute into (**??**):

$$E \;=\; c^2 A(R_E + h)\cdot A(R_E + h)^{-1/2}.$$

Simplify:

$$E \;=\; c^2\,A(R_E + h)^{1/2}.$$

Apply the normalisation at a general r along the worldline ($\mathrm{d}r/\mathrm{d}\tau \neq 0$):

$$-c^2 A(r)\left(\frac{\mathrm{d}t}{\mathrm{d}\tau}\right)^2 \;+\; A(r)^{-1}\left(\frac{\mathrm{d}r}{\mathrm{d}\tau}\right)^2 \;=\; -c^2.$$

Eliminate $\mathrm{d}t/\mathrm{d}\tau$ using $c^2 A(r)\,\mathrm{d}t/\mathrm{d}\tau = E$:

$$-\frac{E^2}{c^2 A(r)} \;+\; A(r)^{-1}\left(\frac{\mathrm{d}r}{\mathrm{d}\tau}\right)^2 \;=\; -c^2.$$

Multiply through by $-A(r)/2$:

$$\frac{1}{2}\left(\frac{\mathrm{d}r}{\mathrm{d}\tau}\right)^2 = \frac{E^2}{2c^2} - \frac{c^2 A(r)}{2}.$$

Insert $E^2 = c^4 A(R_E + h)$:

$$\frac{1}{2}\left(\frac{\mathrm{d}r}{\mathrm{d}\tau}\right)^2 = \frac{c^2}{2}\big[A(R_E + h) - A(r)\big].$$

Substitute $A(r) = 1 - r_s/r$ and cancel the 1's:

$$\frac{1}{2}\left(\frac{\mathrm{d}r}{\mathrm{d}\tau}\right)^2 = \frac{c^2}{2}\left[\frac{r_s}{r} - \frac{r_s}{R_E + h}\right].$$

Factor out r_s:

$$\frac{1}{2}\left(\frac{\mathrm{d}r}{\mathrm{d}\tau}\right)^2 = \frac{c^2 r_s}{2}\left(\frac{1}{r} - \frac{1}{R_E + h}\right).$$

Use $r_s = 2GM/c^2$:

$$\frac{1}{2}\left(\frac{\mathrm{d}r}{\mathrm{d}\tau}\right)^2 = GM\left(\frac{1}{r} - \frac{1}{R_E + h}\right).$$

Evaluate at impact, $r = R_E$:

$$\boxed{\frac{1}{2}\left(\frac{\mathrm{d}r}{\mathrm{d}\tau}\right)^2_{R_E} = GM\left(\frac{1}{R_E} - \frac{1}{R_E + h}\right).} \tag{7}$$

Newtonian limit, expand to first order in h/R_E:

$$\frac{1}{R_E} - \frac{1}{R_E + h} = \frac{h}{R_E(R_E + h)} \approx \frac{h}{R_E^2}.$$

Insert into the boxed result:

$$\tfrac{1}{2}v^2 \approx \frac{GMh}{R_E^2} = gh.$$

Hence $\boxed{v = \sqrt{2gh}.}$

Question 2 — Gravitational wave emission and polarisation

Problem. (a) Identify which of four mass distributions emit GWs. (b) Compute $h_{xx}, h_{yy}, h_{xy}, h_{yx}$ from a rigid dumbbell rotating in the xy-plane, evaluated at large z. (c) Find the displacements of four free test masses, sketch over a cycle, identify polarisation. (d) Discuss polarisation, interference, refraction, diffraction for GR vs EM.

(a) Emission criteria

Linearised GR: leading radiation is mass-quadrupole. A source emits classical GWs iff its mass quadrupole moment $I_{ij} = \int \rho(x_i x_j - \frac{1}{3}\delta_{ij} r^2)\, \mathrm{d}^3x$ has a non-zero *third* time derivative.

(i) Motionless point particle. I_{ij} constant. $\boxed{\text{No emission.}}$

(ii) Rotating sphere (axisymmetric, constant density). Quadrupole is invariant under the rotation. $\dddot{I}_{ij} = 0$. $\boxed{\text{No emission.}}$

(iii) Disc spinning about its symmetry axis. Same axisymmetry: quadrupole invariant. $\dddot{I}_{ij} = 0$. $\boxed{\text{No emission.}}$

(iv) Tilted, precessing disc ($\alpha \neq 0$, precession rate ω). Tilt breaks axisymmetry about z; precession gives a time-varying quadrupole. $\dddot{I}_{ij} \neq 0$. $\boxed{\text{Emits GWs at } 2\omega \text{ (and } \omega \text{ if } \alpha \neq \pi/2).}$

(b) Metric perturbation from a rotating dumbbell

Two masses M at separation $2l$ rotating in the xy-plane at ω. Positions:

$$\boldsymbol{x}_{1,2}(t) \;=\; \pm l\big(\cos\omega t,\, \sin\omega t,\, 0\big).$$

Mass quadrupole (trace-included form):

$$I_{ij}(t) \;=\; \sum_{n=1,2} M\, x_n^i\, x_n^j \;=\; 2Ml^2\, \hat{e}_i(t)\, \hat{e}_j(t), \qquad \hat{e} = (\cos\omega t, \sin\omega t, 0).$$

Compute I_{xx} from $\hat{e}_x^2 = \cos^2\omega t$:

$$I_{xx} \;=\; 2Ml^2 \cos^2\omega t.$$

Apply the double-angle identity $2\cos^2\theta = 1 + \cos 2\theta$:

$$I_{xx} \;=\; Ml^2(1 + \cos 2\omega t).$$

Similarly $I_{yy} = 2Ml^2 \sin^2\omega t$ and $2\sin^2\theta = 1 - \cos 2\theta$:

$$I_{yy} \;=\; Ml^2(1 - \cos 2\omega t).$$

Off-diagonal via $2\sin\theta\cos\theta = \sin 2\theta$:

$$I_{xy} \;=\; I_{yx} \;=\; Ml^2 \sin 2\omega t.$$

Quadrupole formula (TT gauge), retarded at $\boldsymbol{r}$:

$$h_{ij}^{TT}(\boldsymbol{r}, t) \;=\; \frac{2G}{c^4 r}\, \ddot{I}_{ij}^{TT}\Big(t - \frac{r}{c}\Big).$$

Observer on the $+\hat{\boldsymbol{z}}$ axis at $\boldsymbol{r} = z\hat{\boldsymbol{z}}$: wave propagates along $\hat{\boldsymbol{z}}$, transverse plane is xy. The TT projector keeps the symmetric, traceless part of I_{kl} in the xy-plane:

$$I_{xx}^{TT} \;=\; \tfrac{1}{2}(I_{xx} - I_{yy}).$$

Insert the components:

$$I_{xx}^{TT} \;=\; \tfrac{1}{2}[Ml^2(1 + \cos 2\omega t) - Ml^2(1 - \cos 2\omega t)] \;=\; Ml^2 \cos 2\omega t.$$

By symmetry $I_{yy}^{TT} = -I_{xx}^{TT}$ and $I_{xy}^{TT} = I_{xy}$:

$$I_{yy}^{TT} \;=\; -Ml^2 \cos 2\omega t, \qquad I_{xy}^{TT} \;=\; Ml^2 \sin 2\omega t.$$

Differentiate twice using $\mathrm{d}^2(\cos 2\omega t)/\mathrm{d}t^2 = -4\omega^2 \cos 2\omega t$:

$$\ddot{I}_{xx}^{TT}(t_r) \;=\; -4Ml^2\omega^2 \cos[2\omega(t - z/c)].$$
$$\ddot{I}_{xy}^{TT}(t_r) \;=\; -4Ml^2\omega^2 \sin[2\omega(t - z/c)].$$

Substitute into the quadrupole formula at $r = z$:

$$\boxed{\begin{aligned} h_{xx}(z,t) = -h_{yy}(z,t) \;&=\; -\frac{8GMl^2\omega^2}{c^4 z} \cos[2\omega(t - z/c)], \\ h_{xy}(z,t) = h_{yx}(z,t) \;&=\; -\frac{8GMl^2\omega^2}{c^4 z} \sin[2\omega(t - z/c)]. \end{aligned}} \tag{8}$$

GW frequency $= 2\omega$ (mass quadrupole). Define amplitude $h_0 \equiv 8GMl^2\omega^2/(c^4 z)$.

(c) Free-mass displacements and polarisation

Geodesic deviation in TT gauge for masses at rest at ξ^i:

$$\ddot{\xi}^i \;=\; \tfrac{1}{2}\ddot{h}^i{}_j(t)\,\xi^j.$$

Integrate twice (steady-state, drop transients):

$$\delta\xi^i(t) \;=\; \tfrac{1}{2}h^i{}_j(t)\,\xi^j_{(0)}.$$

Mass at $(a,0,0)$:

$$\delta\xi_x \;=\; \tfrac{a}{2}h_{xx}(t), \qquad \delta\xi_y \;=\; \tfrac{a}{2}h_{xy}(t), \qquad \delta\xi_z \;=\; 0.$$

Mass at $(-a,0,0)$: opposite sign of above. Mass at $(0,a,0)$:

$$\delta\xi_x \;=\; \tfrac{a}{2}h_{xy}(t), \qquad \delta\xi_y \;=\; \tfrac{a}{2}h_{yy}(t) \;=\; -\tfrac{a}{2}h_{xx}(t).$$

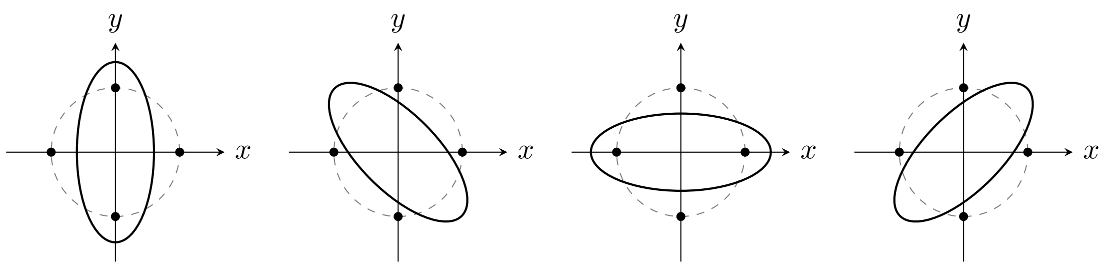
Mass at $(0,-a,0)$: opposite sign. All four masses move only in the xy-plane: $\xi_z = 0$.

Polarisation. $h_{xx} = -h_{yy}$ (the $+$ amplitude) and h_{xy} (the $\times$ amplitude) have equal magnitude h_0 and are 90° out of phase (cos vs sin). $\boxed{\text{Equal-amplitude superposition of } + \text{ and } \times \text{ at relative phase } \pi/2 = \text{circularly}}$

Sketch. Initial square at $(a,0),(0,a),(-a,0),(0,-a)$. Phase $\Phi = 2\omega t$:

- $\Phi = 0$: $h_{xx} < 0$, $h_{xy} = 0 \Rightarrow$ vertical stretch, horizontal squeeze (square $\to$ tall ellipse).
- $\Phi = \pi/2$: $h_{xx} = 0$, $h_{xy} < 0 \Rightarrow$ stretch along NW–SE diagonal (ellipse rotated $+45°$ from vertical).
- $\Phi = \pi$: $h_{xx} > 0 \Rightarrow$ horizontal stretch (wide ellipse).
- $\Phi = 3\pi/2$: $h_{xy} > 0 \Rightarrow$ stretch along NE–SW diagonal (ellipse rotated $+135°$ from vertical).

The pattern is a single ellipse rotating uniformly counter-clockwise (viewed from $+z$ looking down) at rate ω, following the source rotation.



$0 = 0_0$ $\quad$ $90 = 90_1$ $\quad$ $180 = 180_2$ $\quad$ $270 = 270_3$

$\Phi_0 = 0,\ \Phi_1 = \pi/2,\ \Phi_2 = \pi,\ \Phi_3 = 3\pi/2$. Dashed circle: original positions. Ellipse rotates CCW at ω (one full π of orientation per GW half-period π/ω).

(d) GR vs EM: polarisation, interference, refraction, diffraction

(i) Polarisation. Both wave fields are transverse. EM has 2 spin-1 modes ($\hat{\boldsymbol{e}}_x, \hat{\boldsymbol{e}}_y$, transforming under 90° rotations). GR has 2 spin-2 modes ($+$ and $\times$, transforming under 45° rotations as visualised above). Plausible: yes; *detected* by LIGO via the strain pattern.

(ii) Interference. Linearised GR is a linear theory in vacuum: $h_{\mu\nu} = h^{(1)}_{\mu\nu} + h^{(2)}_{\mu\nu}$ obeys superposition. Coherent two-source interference is therefore possible in principle. Plausibility: requires two phase-coherent astrophysical sources at the same frequency observed simultaneously — not realised astrophysically; lensing-induced self-interference (wave optics in strong lensing) is plausible at ∼mHz with LISA.

(iii) Refraction. EM refraction comes from microscopic charges polarising in matter. GR couples to stress–energy of *everything* but with strength G/c^4, giving an index $n_{\mathrm{GR}} - 1 \sim G\rho/(\omega^2 c^2)$. For nuclear-density matter and kHz frequencies, $n_{\mathrm{GR}} - 1 \sim 10^{-20}$. Possible in principle, utterly negligible in practice. $\boxed{\text{No measurable refractive medium for GWs.}}$

(iv) Diffraction. Standard wave phenomenon: GW of wavelength $\lambda_{\mathrm{GW}} = c/f$ around an obstacle of size D shows diffraction when $D \lesssim \lambda_{\mathrm{GW}}$. Astrophysical lensing in the wave-optics regime ($\lambda_{\mathrm{GW}} \gtrsim r_s$ of the lens) does diffract GWs and is a target of LISA / pulsar timing. Plausible *and* potentially observable for solar-mass lenses at μHz.

Summary. Polarisation and diffraction are real and observable. Interference is mathematically possible but astrophysically rare. Refraction is in principle possible but quantitatively negligible.

Question 3 — Gyroscope precession in Schwarzschild orbit

Problem. (a) Show $u^0 = \mathrm{d}t/\mathrm{d}\tau = (1 - 3GM/rc^2)^{-1/2}$ for a circular geodesic. (b) For a spin 4-vector s initially radial at $t = 0$, parallel-transport gives geodetic precession at rate $\Omega_r = \Omega/u^0$. (c) Numerical orbital periods (coordinate vs proper) for $M = 8.2 \times 10^{36}$ kg, $R = 200R_s$. (d) Will a 1° beam still hit Earth after one orbit; is path-curvature important?

(a) Time dilation factor on a circular orbit

Schwarzschild equatorial circular orbit: $\theta = \pi/2$, $r = \text{const}$, $\dot{r} = \dot{\theta} = 0$. Apply normalisation $g_{\mu\nu}u^\mu u^\nu = -c^2$:

$$-c^2 A(r)(u^0)^2 \;+\; r^2(u^\phi)^2 \;=\; -c^2, \qquad A(r) \equiv 1 - \frac{r_s}{r} = 1 - \frac{2GM}{rc^2}.$$

Radial geodesic equation, with $\mathrm{d}u^r/\mathrm{d}\tau = 0$:

$$\Gamma^r_{00}(u^0)^2 \;+\; \Gamma^r_{\phi\phi}(u^\phi)^2 \;=\; 0.$$

Relevant Christoffel symbols (Schwarzschild, $\theta = \pi/2$):

$$\Gamma^r_{00} \;=\; \frac{c^2 r_s}{2r^2}A(r), \qquad \Gamma^r_{\phi\phi} \;=\; -rA(r).$$

Substitute into the geodesic equation:

$$\frac{c^2 r_s}{2r^2}A(r)(u^0)^2 \;-\; rA(r)(u^\phi)^2 \;=\; 0.$$

Cancel the common $A(r)$:

$$\frac{c^2 r_s}{2r^2}(u^0)^2 \;=\; r(u^\phi)^2.$$

Solve for $(u^\phi/u^0)^2$:

$$\left(\frac{u^\phi}{u^0}\right)^2 = \frac{c^2 r_s}{2r^3}.$$

Insert $r_s = 2GM/c^2$ to recover Kepler:

$$\Omega^2 \equiv \left(\frac{\mathrm{d}\phi}{\mathrm{d}t}\right)^2 = \frac{GM}{r^3}. \tag{9}$$

Rewrite the normalisation using $u^\phi = \Omega\, u^0$:

$$-c^2 A(r)(u^0)^2 + r^2\Omega^2(u^0)^2 = -c^2.$$

Factor $(u^0)^2$:

$$(u^0)^2\big[c^2 A(r) - r^2\Omega^2\big] = c^2.$$

Compute the bracket: expand $c^2A(r) = c^2 - 2GM/r$ and substitute $r^2\Omega^2 = GM/r$:

$$c^2A(r) - r^2\Omega^2 = c^2 - \frac{2GM}{r} - \frac{GM}{r}.$$

Combine the $1/r$ terms:

$$c^2A(r) - r^2\Omega^2 = c^2\left(1 - \frac{3GM}{rc^2}\right).$$

Solve for u^0:

$$\boxed{u^0 = \frac{\mathrm{d}t}{\mathrm{d}\tau} = \left(1 - \frac{3GM}{rc^2}\right)^{-1/2}.} \tag{10}$$

Symbols: t Schwarzschild coordinate time (asymptotic observer); τ proper time of the orbiting body; r areal radius; M central mass; G, c Newton/light constants; u^0 time-component of 4-velocity.

Sanity check. At the photon sphere $r = 3GM/c^2 = 1.5r_s$, $u^0 \to \infty$ — timelike circular orbits cease to exist and become null (photon) geodesics.

(b) Parallel transport of the spin

Parallel transport along worldline u^μ:

$$\frac{\mathrm{d}s^\mu}{\mathrm{d}\tau} + \Gamma^\mu_{\alpha\beta}\, u^\alpha s^\beta = 0. \tag{11}$$

Spin orthogonality at $\tau = 0$: $u_\mu s^\mu = 0$. Initially radial: $s^\mu(0) = (0, s^r, 0, 0)$. Equatorial orbit, $\theta = \pi/2$.

Required Christoffels (Schwarzschild, $\theta = \pi/2$):

$$\Gamma^t_{tr} = \frac{r_s}{2r^2 A(r)}, \qquad \Gamma^r_{tt} = \frac{c^2 r_s}{2r^2} A(r).$$

$$\Gamma^r_{\phi\phi} = -rA(r), \qquad \Gamma^\phi_{r\phi} = \frac{1}{r}, \qquad \Gamma^\theta_{\phi\phi} = 0 \ (\sin\cos|_{\pi/2} = 0).$$

4-velocity has only u^t, u^ϕ non-zero. The θ-component of (**??**) closes trivially: no Christoffels mix θ into the (t, r, ϕ) block at $\theta = \pi/2$, so $s^\theta \equiv 0$. $\boxed{s^\theta(t) = 0.}$

Convert $\mathrm{d}/\mathrm{d}\tau = u^0\,\mathrm{d}/\mathrm{d}t$. The (t, r, ϕ) components of (**??**):

$$u^0\frac{\mathrm{d}s^t}{\mathrm{d}t} \;+\; \Gamma^t_{tr}(u^t s^r + u^r s^t) \;+\; \Gamma^t_{t\phi}(\cdot) \;=\; 0.$$

With $u^r = 0$ and $\Gamma^t_{t\phi} = 0$:

$$\frac{\mathrm{d}s^t}{\mathrm{d}t} \;=\; -\Gamma^t_{tr}\, s^r.$$

Radial component:

$$u^0\frac{\mathrm{d}s^r}{\mathrm{d}t} \;+\; \Gamma^r_{tt}\, u^t s^t \;+\; \Gamma^r_{\phi\phi}\, u^\phi s^\phi \;=\; 0.$$

Divide by u^0 and use $u^t = u^0$, $u^\phi = \Omega u^0$:

$$\frac{\mathrm{d}s^r}{\mathrm{d}t} \;=\; -\Gamma^r_{tt}\, s^t \;-\; \Gamma^r_{\phi\phi}\,\Omega\, s^\phi.$$

Azimuthal component (only $\Gamma^\phi_{r\phi}$ survives):

$$u^0\frac{\mathrm{d}s^\phi}{\mathrm{d}t} \;+\; \Gamma^\phi_{r\phi}(u^r s^\phi + u^\phi s^r) \;=\; 0.$$

With $u^r = 0$:

$$\frac{\mathrm{d}s^\phi}{\mathrm{d}t} \;=\; -\frac{\Omega}{r}\, s^r.$$

Eliminate s^t via orthogonality $u_\mu s^\mu = 0$, i.e. $-c^2 A(r)\, u^0 s^t + r^2 u^\phi s^\phi = 0$:

$$s^t \;=\; \frac{r^2\Omega}{c^2 A(r)}\, s^\phi.$$

Substitute into the $\mathrm{d}s^r/\mathrm{d}t$ equation:

$$\frac{\mathrm{d}s^r}{\mathrm{d}t} \;=\; \left[-\Gamma^r_{tt}\frac{r^2\Omega}{c^2 A(r)} - \Gamma^r_{\phi\phi}\,\Omega\right] s^\phi.$$

Evaluate the first term using $\Gamma^r_{tt} = \frac{c^2 r_s}{2r^2} A(r)$:

$$-\Gamma^r_{tt}\frac{r^2\Omega}{c^2 A(r)} \;=\; -\frac{c^2 r_s}{2r^2} A(r)\cdot\frac{r^2\Omega}{c^2 A(r)} \;=\; -\frac{r_s\Omega}{2}.$$

Evaluate the second term using $\Gamma^r_{\phi\phi} = -rA(r)$:

$$-\Gamma^r_{\phi\phi}\,\Omega \;=\; rA(r)\,\Omega \;=\; (r - r_s)\Omega.$$

Sum the two contributions:

$$\frac{\mathrm{d}s^r}{\mathrm{d}t} \;=\; \Omega\left(r - r_s - \frac{r_s}{2}\right) s^\phi.$$

Combine the r_s terms:

$$\frac{\mathrm{d}s^r}{\mathrm{d}t} \;=\; \Omega\left(r - \frac{3r_s}{2}\right) s^\phi.$$

Factor r:

$$\frac{\mathrm{d}s^r}{\mathrm{d}t} \;=\; \Omega\, r\left(1 - \frac{3r_s}{2r}\right) s^\phi.$$

Substitute $r_s = 2GM/c^2$ to recognise $(u^0)^{-2}$:

$$\frac{\mathrm{d}s^r}{\mathrm{d}t} \;=\; \Omega\, r\left(1 - \frac{3GM}{rc^2}\right) s^\phi.$$

Hence, using $(u^0)^{-2} = 1 - 3GM/(rc^2)$:

$$\frac{\mathrm{d}s^r}{\mathrm{d}t} \;=\; \frac{\Omega\, r}{(u^0)^2}\, s^\phi. \tag{12}$$

Pair with $\mathrm{d}s^\phi/\mathrm{d}t = -(\Omega/r)\, s^r$:

$$\frac{\mathrm{d}s^\phi}{\mathrm{d}t} \;=\; -\frac{\Omega}{r}\, s^r. \tag{13}$$

Differentiate (**??**) in t (note r, u^0, Ω are constants on the orbit):

$$\frac{\mathrm{d}^2 s^r}{\mathrm{d}t^2} \;=\; \frac{\Omega r}{(u^0)^2}\,\frac{\mathrm{d}s^\phi}{\mathrm{d}t}.$$

Substitute (**??**) for $\mathrm{d}s^\phi/\mathrm{d}t$:

$$\frac{\mathrm{d}^2 s^r}{\mathrm{d}t^2} \;=\; \frac{\Omega r}{(u^0)^2}\left(-\frac{\Omega}{r}\, s^r\right) \;=\; -\frac{\Omega^2}{(u^0)^2}\, s^r.$$

Identify the angular frequency of $s^r(t)$:

$$\boxed{\Omega_r \;=\; \frac{\Omega}{u^0} \;=\; \Omega\sqrt{1 - \frac{3GM}{rc^2}}.} \tag{14}$$

The spin's radial component oscillates at Ω_r with respect to the asymptotic frame. Geodetic precession per orbit is the difference $\Omega - \Omega_r$ multiplied by the orbital period (return-to-radial mismatch).

Physical picture. In one coordinate orbit the radial axis returns; the spin returns to a direction lagging by $2\pi(1 - \Omega_r/\Omega) = 2\pi(1 - 1/u^0)$ — the de Sitter geodetic angle.

(c) Orbital periods of the Rocinante

Data: $M = 8.2 \times 10^{36}$ kg, $R = 200R_s$.
Schwarzschild radius from $R_s = 2GM/c^2$:

$$R_s \;=\; \frac{2 \cdot 6.674 \times 10^{-11} \cdot 8.2 \times 10^{36}}{(3 \times 10^8)^2}\ \mathrm{m}.$$

Evaluate numerator:

$$2 \cdot 6.674 \times 10^{-11} \cdot 8.2 \times 10^{36} \;=\; 1.0945 \times 10^{27}\,\mathrm{m}^3\,\mathrm{s}^{-2}.$$

Divide by $c^2 = 9 \times 10^{16}\,\mathrm{m}^2\,\mathrm{s}^{-2}$:

$$R_s \;=\; 1.216 \times 10^{10}\,\mathrm{m}.$$

Orbital radius:

$$R \;=\; 200R_s \;=\; 2.432 \times 10^{12}\,\mathrm{m}.$$

Coordinate angular frequency from Kepler (**??**):

$$\Omega \;=\; \sqrt{\frac{GM}{R^3}}.$$

Compute R^3:

$$R^3 \;=\; (2.432\times 10^{12})^3 \;=\; 1.439\times 10^{37}\,\mathrm{m}^3.$$

Compute GM:

$$GM \;=\; 6.674\times 10^{-11}\cdot 8.2\times 10^{36} \;=\; 5.473\times 10^{26}\,\mathrm{m}^3\,\mathrm{s}^{-2}.$$

Divide:

$$\frac{GM}{R^3} \;=\; \frac{5.473\times 10^{26}}{1.439\times 10^{37}} \;=\; 3.804\times 10^{-11}\,\mathrm{s}^{-2}.$$

Take square root:

$$\Omega \;=\; 6.17\times 10^{-6}\,\mathrm{rad\,s}^{-1}.$$

Coordinate period (distant observer):

$$T_\infty \;=\; \frac{2\pi}{\Omega} \;=\; \frac{6.283}{6.17\times 10^{-6}}\,\mathrm{s}.$$

Evaluate:

$$\boxed{T_\infty \;\approx\; 1.018\times 10^{6}\,\mathrm{s} \;\approx\; 11.8\,\mathrm{days}.} \tag{15}$$

Crew period (proper time per orbit) from $\mathrm{d}\tau = \mathrm{d}t/u^0$. Use $r = 200R_s$ so $3GM/(rc^2) = 3R_s/(2r) = 3/(400)$:

$$\frac{3GM}{rc^2} \;=\; 7.5\times 10^{-3}.$$

Compute $1/u^0$:

$$\frac{1}{u^0} \;=\; \sqrt{1-7.5\times 10^{-3}} \;=\; 0.99624.$$

Proper period:

$$T_{\mathrm{crew}} \;=\; \frac{T_\infty}{u^0} \;=\; 1.018\times 10^{6}\cdot 0.99624\,\mathrm{s}.$$

Evaluate:

$$\boxed{T_{\mathrm{crew}} \;\approx\; 1.014\times 10^{6}\,\mathrm{s}, \quad T_\infty - T_{\mathrm{crew}} \;\approx\; 3.8\times 10^{3}\,\mathrm{s}.} \tag{16}$$

Why different. Two effects, both captured in u^0: gravitational time dilation ($\sqrt{A(r)}$) and special-relativistic time dilation from orbital speed ($r\Omega/c$). Together they give $(u^0)^{-1} = \sqrt{1-3GM/rc^2}$, so the crew's clock runs slow.

(d) Pointing after one orbit and curvature distortion

Initial spin direction: along the radial line to Earth at t_0. After one coordinate orbit ($\Delta t = T_\infty$), the radial direction returns, but s rotates at Ω_r in the local frame.

Geodetic angle per coordinate orbit:

$$\Delta\phi_{\mathrm{geo}} \;=\; 2\pi\left(1-\frac{\Omega_r}{\Omega}\right) \;=\; 2\pi\left(1-\frac{1}{u^0}\right).$$

Substitute $1/u^0 = \sqrt{1 - 3GM/(Rc^2)}$:

$$\Delta\phi_{\text{geo}} \;=\; 2\pi\left(1 - \sqrt{1 - \frac{3GM}{Rc^2}}\right).$$

Insert the numerical value $1/u^0 = 0.99624$:

$$\Delta\phi_{\text{geo}} \;=\; 2\pi(1 - 0.99624) \;=\; 2\pi \cdot 3.76 \times 10^{-3}.$$

Evaluate in radians:

$$\Delta\phi_{\text{geo}} \;=\; 2.36 \times 10^{-2}\,\text{rad}.$$

Convert to degrees ($\times 180/\pi$):

$$\boxed{\Delta\phi_{\text{geo}} \;\approx\; 1.35^\circ \,/\, \text{orbit}.} \tag{17}$$

Beam angular diameter 1° gives a half-angle of 0.5°. Compare:

$$\Delta\phi_{\text{geo}} \;=\; 1.35^\circ \;>\; 0.5^\circ.$$

$\boxed{\text{The beam misses Earth: the gyroscope axis has precessed by more than the beam half-angle.}}$

Path curvature of the beam. Schwarzschild deflection of a light ray passing at impact parameter b is $\delta\alpha = 2R_s/b$ (small-deflection limit). For a beam emitted radially at $r = 200R_s$ travelling outward, the impact parameter relative to M is $\sim r$ initially and grows. Order of magnitude:

$$\delta\alpha \;\lesssim\; \frac{2R_s}{R} \;=\; \frac{2}{200} \;=\; 1 \times 10^{-2}\,\text{rad} \;\approx\; 0.57^\circ.$$

This is comparable to the geodetic precession; *however*, the beam is launched outward and away from M, not grazing past it, so the actual deflection on the outward leg is far smaller (nearly zero curvature for radial null geodesics in Schwarzschild). $\boxed{\text{Path-bending of the radially-launched beam is negligible compared to g}}$

The dominant pointing error is geodetic precession of the gyroscope, not bending of the photon path.

Question 4 — FRW cosmology, equations of state, age of the Universe

Problem. (a) Best-fit FRW metric for our Universe; how it differs from the general FRW. (b) Derive the continuity equation $\dot\rho + 3(\rho + P/c^2)\dot R/R = 0$ and show its equivalence to the first law of thermodynamics. (c) Identify three equations of state: $P = -\rho c^2$, $P = \frac{1}{3}\rho c^2$, $P = 0$. (d) Express the age of the Universe as $\int \mathrm{d}R/(R\,H)$ and prove that including S1 makes the Universe older. (e) Comment on differences from the real Universe.

(a) FRW metric for the Universe

Cosmological observations (CMB, BAO, large-scale isotropy) are consistent with spatial flatness $k = 0$:

$$\boxed{\mathrm{d}s^2 \;=\; -c^2\,\mathrm{d}t^2 \;+\; R^2(t)\big[\mathrm{d}r^2 + r^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2)\big].} \tag{18}$$

General FRW allows $k = 0, \pm 1$:

$$\mathrm{d}s^2 \;=\; -c^2\,\mathrm{d}t^2 \;+\; R^2(t)\left[\frac{\mathrm{d}r^2}{1 - kr^2} + r^2\,\mathrm{d}\Omega^2\right].$$

Symbols: t cosmic (proper) time of comoving observers; $R(t)$ scale factor with dimensions of length (or dimensionless if absorbed into r); (r, θ, ϕ) comoving spherical coordinates; k spatial-curvature constant. Our Universe corresponds to $k = 0$ to within current observational error.

(b) Continuity equation and first law

Local energy–momentum conservation $\nabla_\mu T^{\mu\nu} = 0$ for a perfect fluid at rest in the FRW frame. The $\nu = 0$ component reads

$$\partial_\mu T^{\mu 0} \;+\; \Gamma^{\mu}_{\mu\alpha} T^{\alpha 0} \;+\; \Gamma^{0}_{\alpha\beta} T^{\alpha\beta} \;=\; 0.$$

For flat FRW, $\Gamma^{\mu}_{\mu 0} = 3\dot{R}/R$ (volume expansion of the spatial slice) and $\Gamma^{0}_{ij} \propto R\dot{R}\,\delta_{ij}$. Substituting the comoving $T^{\mu\nu}$ of a perfect fluid (energy density ρc^2, isotropic pressure P):

$$\dot{\rho}c^2 \;+\; 3\frac{\dot{R}}{R}\rho c^2 \;+\; 3\frac{\dot{R}}{R}P \;=\; 0.$$

Divide through by c^2:

$$\boxed{\dot{\rho} \;+\; 3(\rho + P/c^2)\frac{\dot{R}}{R} \;=\; 0.} \tag{19}$$

Symbols: overdot = d/dt; ρ rest-frame mass-energy density; P rest-frame pressure; R scale factor.

First-law equivalence. Comoving volume $V \propto R^3$, energy $E = \rho c^2 V$. Differentiate in t (product rule):

$$\frac{\mathrm{d}E}{\mathrm{d}t} \;=\; \dot{\rho}\,c^2 V \;+\; \rho c^2\,\dot{V}.$$

Differentiate $V \propto R^3$ to give $\dot{V} = 3V\,\dot{R}/R$:

$$\frac{\mathrm{d}E}{\mathrm{d}t} \;=\; \dot{\rho}\,c^2 V \;+\; 3\rho c^2 V\,\frac{\dot{R}}{R}.$$

Solve (**??**) for $\dot{\rho}c^2$:

$$\dot{\rho}\,c^2 \;=\; -3(\rho c^2 + P)\,\frac{\dot{R}}{R}.$$

Substitute and expand:

$$\frac{\mathrm{d}E}{\mathrm{d}t} \;=\; -3(\rho c^2 + P)V\,\frac{\dot{R}}{R} \;+\; 3\rho c^2 V\,\frac{\dot{R}}{R}.$$

Cancel the $\rho c^2 V\dot{R}/R$ terms:

$$\frac{\mathrm{d}E}{\mathrm{d}t} \;=\; -3PV\,\frac{\dot{R}}{R}.$$

Recognise $3V\dot{R}/R = \dot{V}$:

$$\frac{\mathrm{d}E}{\mathrm{d}t} \;=\; -P\,\frac{\mathrm{d}V}{\mathrm{d}t}.$$

Multiply by dt:

$$\boxed{\mathrm{d}E \;=\; -P\,\mathrm{d}V.} \tag{20}$$

First law of thermodynamics for an adiabatic (d$Q = 0$) expansion of a comoving volume.

(c) Three equations of state

S1: $P = -\rho c^2$**.** Substitute into (**??**) with $\rho + P/c^2 = 0$:

$$\dot{\rho} \;=\; 0.$$

Density is constant during expansion. $\boxed{\text{Cosmological constant / vacuum energy / dark energy } (\Lambda)\text{; drives accelerate}}$

S2: $P = \frac{1}{3}\rho c^2$**.** Substitute $P/c^2 = \rho/3$ into (**??**):

$$\dot\rho \;+\; 4\rho\,\frac{\dot R}{R} \;=\; 0.$$

Separate variables:

$$\frac{\dot\rho}{\rho} \;=\; -4\,\frac{\dot R}{R}.$$

Integrate to obtain $\rho \propto R^{-4}$. $\boxed{\text{Radiation-dominated era (photons, ultrarelativistic species). Trace-free } T^{\mu}{}_{\mu} = 0.}$

S3: $P = 0$**.** Substitute into (**??**):

$$\dot\rho \;+\; 3\rho\,\frac{\dot R}{R} \;=\; 0.$$

Separate and integrate:

$$\rho \;\propto\; R^{-3}.$$

$\boxed{\text{Matter-dominated era (cold pressureless dust: baryons + cold dark matter).}}$

(d) Age of the Universe and the effect of S1

Hubble parameter $H = \dot R/R$:

$$\mathrm{d}t \;=\; \frac{\mathrm{d}R}{R\,H(R)}.$$

Integrate from $R = 0$ to $R = R_0$ (today):

$$\boxed{t_0 \;=\; \int_0^{R_0} \frac{\mathrm{d}R}{R\,H(R)}.} \tag{21}$$

Phase-by-phase $H(R)$**.** Friedmann ($k = 0$, $\Lambda = 0$, density evolved per equation of state):

$$H^2 \;=\; \frac{8\pi G}{3}\rho(R).$$

For matter $\rho_m = \rho_{m,0}(R_0/R)^3 \Rightarrow H_m \propto R^{-3/2}$. For radiation $\rho_r = \rho_{r,0}(R_0/R)^4 \Rightarrow H_r \propto R^{-2}$. For dark energy ρ_Λ constant $\Rightarrow H_\Lambda$ constant.

Universe *without* S1. Phases S2 then S3. Pick equality scale factor $R_{\rm eq}$ at which radiation–matter densities cross, then today's R_0.

Radiation phase contribution (from 0 to $R_{\rm eq}$). Use $H(R) = H_{\rm eq}(R_{\rm eq}/R)^2$:

$$t_r \;=\; \int_0^{R_{\rm eq}} \frac{\mathrm{d}R}{R\,H_{\rm eq}(R_{\rm eq}/R)^2} \;=\; \frac{1}{H_{\rm eq}\,R_{\rm eq}^2}\int_0^{R_{\rm eq}} R\,\mathrm{d}R.$$

Evaluate the integral:

$$t_r \;=\; \frac{1}{H_{\rm eq}\,R_{\rm eq}^2}\cdot\frac{R_{\rm eq}^2}{2} \;=\; \frac{1}{2H_{\rm eq}}.$$

Matter phase (from $R_{\rm eq}$ to R_0). Use $H(R) = H_{\rm eq}(R_{\rm eq}/R)^{3/2}$:

$$t_m \;=\; \int_{R_{\rm eq}}^{R_0} \frac{\mathrm{d}R}{R\,H_{\rm eq}(R_{\rm eq}/R)^{3/2}} \;=\; \frac{1}{H_{\rm eq}\,R_{\rm eq}^{3/2}}\int_{R_{\rm eq}}^{R_0} R^{1/2}\,\mathrm{d}R.$$

Evaluate the power-law integral:

$$t_m \;=\; \frac{1}{H_{\rm eq}\, R_{\rm eq}^{3/2}} \cdot \frac{2}{3}\Big[R_0^{3/2} - R_{\rm eq}^{3/2}\Big].$$

Simplify:

$$t_m \;=\; \frac{2}{3H_{\rm eq}}\left[\left(\frac{R_0}{R_{\rm eq}}\right)^{3/2} - 1\right].$$

Sum:

$$t_0^{(\text{no S1})} \;=\; t_r + t_m \;=\; \frac{1}{2H_{\rm eq}} + \frac{2}{3H_{\rm eq}}\left[\left(\frac{R_0}{R_{\rm eq}}\right)^{3/2} - 1\right].$$

Universe *with* S1. Insert dark-energy phase first ($R \in [0, R_1]$) before S2 takes over at R_1. With ρ_Λ constant, the Friedmann equation gives $H(R) = H_\Lambda$ on $[0, R_1]$:

$$t_\Lambda \;=\; \int_{R_{\rm min}}^{R_1} \frac{\mathrm{d}R}{R\, H_\Lambda}.$$

Evaluate the logarithmic integral:

$$t_\Lambda \;=\; \frac{1}{H_\Lambda}\, \ln\!\left(\frac{R_1}{R_{\rm min}}\right).$$

Take the limit $R_{\rm min} \to 0^+$:

$$\lim_{R_{\rm min}\to 0^+} \frac{1}{H_\Lambda}\, \ln\!\left(\frac{R_1}{R_{\rm min}}\right) \;=\; +\infty.$$

The S1 phase is exponential expansion (de Sitter), $R(t) = R_1\, e^{H_\Lambda(t-t_1)}$, so reaching any $R_1 > 0$ from $R \to 0$ takes unbounded coordinate time. Subsequent S2 and S3 phases contribute exactly the same $t_r + t_m$ as in the no-S1 case. Therefore

$$\boxed{t_0^{(\text{with S1})} \;=\; t_\Lambda + t_r + t_m \;>\; t_0^{(\text{no S1})}.} \tag{22}$$

$\boxed{\text{A Universe with S1 is older than one without.}}$

(e) Real Universe versus model

Concrete differences:

1. **Sequence.** The real Universe ordering is roughly S2 (radiation, $0 < t < 50\,\text{kyr}$) $\to$ S3 (matter, dominant from $\sim 50\,\text{kyr}$ to $\sim 9\,\text{Gyr}$) $\to$ S1 (dark-energy, $z \lesssim 0.5$). The exam model has S1 *first*; in reality Λ dominates last.
2. **Inflation.** A separate, much earlier de-Sitter-like phase (energy scale $\sim 10^{16}$ GeV, duration $\sim 10^{-32}$ s) is generally invoked to solve flatness/horizon/monopole problems — this *is* an S1-type stage but at the GUT scale, not at low z.
3. **Mixed components.** The real Universe is at all times a mixture of radiation, matter (baryons + CDM), and dark energy: pure single-fluid phases are an idealisation.
4. **Curvature.** Observations constrain $|\Omega_k| < 0.01$; we take $k = 0$ but a small non-zero k is not excluded.
5. **Neutrino transition.** Neutrinos are radiation while ultrarelativistic, become matter-like once $T < m_\nu c^2/k_B$, smoothly modifying the radiation $\to$ matter transition.

6. **Reheating + recombination.** Phase boundaries are not sharp: reheating at the end of inflation, e^+e^- annihilation, BBN, recombination at $z \approx 1100$ all introduce structure absent from a three-stage piecewise model.

The three-phase model is qualitatively right (each EOS is realised at some epoch) but mis-orders S1 and ignores all transitions and admixtures.

End of solutions.